SQL Project - Zomato Sales Analysis

Project Overview:

This project involves analyzing a Zomato sales database containing data on customers, orders, restaurants, employees, and food items. The goal is to perform various SQL queries to extract and analyze data, providing insights and helping students develop their SQL skills through practical application.

Note: PFA database file, 'zomatodf.sql', run it in your workbench to create the database.

Task Breakdown:

1. List the top 3 customers who have placed the highest number of orders. order by Highest_No_Of_Order desc limit 3;

```
select customer.customer_id ,customer.customer_name,

count(order_detail.order_id) as Highest_No_Of_Order

from customer

join order_detail

on customer.customer_id = order_detail.customer_id

group by customer.customer_id
```

Result Output -

customer_id	customer_name	Highest_No_Of_Order
1001	vaibhav Palkar	11
1008	punam kale	4
1003	sagar amritkar	4

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2. Find the restaurant that has received the highest average rating.

```
select
restaurant_id, restaurant_name,avg(rrating) as Rating
from restaurant
group by restaurant_id ,restaurant_name
order by  Rating desc limit 1;
```

Result Output -

restaurant_id	restaurant_name	Rating
101	hydrabadi spice	4.5

3. List all orders from order_detail that were delivered in under 30 minutes.

```
select (order_id),customer_id,
order_time,delivered_time,
timestampdiff(minute, order_time, delivered_time) as delivery_time_in_minutes
from order_detail
where timestampdiff(minute, order_time, delivered_time) < 30;
```

Result Output -

order_id	customer_id	order_time	delivered_time	delivery_time_in_minutes
7719	1009	28/10/2019 21:36	28/10/2019 21:45	8
7720	1010	12/10/2019 12:36	12/10/2019 13:01	24
7721	1011	14/10/2019 12:36	14/10/2019 13:01	24
7724	1004	25/10/2019 20:01	25/10/2019 20:29	28
7727	1014	28/10/2019 21:36	28/10/2019 21:45	8
7731	1007	21/10/2019 20:30	21/10/2019 20:55	24
7734	1005	12/10/2019 12:36	12/10/2019 13:01	24
7737	1002	23/10/2019 13:35	23/10/2019 14:02	26
7744	1010	23/10/2019 12:36	23/10/2019 13:01	24
7746	1007	21/10/2019 20:30	21/10/2019 20:55	24
7749	1017	14/09/2019 10:23	14/09/2019 10:51	28

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4. Calculate the total revenue generated by each food item. Display food_id, food_name and Total revenue ordered by total revenue in descending order.

```
select foods.food_id,foods.food_name,
sum(order_food.quantity * foods.price_per_unit) as Total_Revenue
from foods join
order_food on foods.food_id = order_food.food_id
group by food_id,food_name
order by Total_Revenue  desc ;
```

Result Output -

food_id	food_name	Total_Revenue
9999416	veg biryani	2520
9999419	mix paratha	1620
9999417	mix veg	900
9999422	veg roll	900
9999411	allo paratha	880
9999420	cold drink	665
9999414	veg thali	660
9999412	veg meal	480
9999418	veg pulav	380
9999421	paneer roll	360
9999415	chicken thali	300
9999413	chapati	240

5. Find the second highest revenue-generating restaurant.

```
with  RevenueCTE as (
select  restaurant.restaurant_id ,restaurant.restaurant_name,
sum(order_food.quantity * foods.price_per_unit) as Tatal_Revenue,
dense_rank() over( order by sum(order_food.quantity * foods.price_per_unit)
desc) as revenue_rank
from restaurant
join  order_food
on restaurant.restaurant_id =order_food.restaurant_id
join foods
on order_food.food_id = foods.food_id
group by restaurant.restaurant_id, restaurant.restaurant_name
)
select * from RevenueCTE
where revenue_rank = 2;
```

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Result Output -

restaurant_id	restaurant_name	Tatal_Revenue	revenue_rank
101	hydrabadi spice	1635	2

6. Find the 5 most popular food items based on the quantity sold.

```
select
foods.food_id,foods.food_name,
sum(order_food.quantity) as Total_Quantity_Sold
from foods
join order_food on foods.food_id = order_food.food_id
group by foods.food_id,foods.food_name
order by  Total_Quantity_Sold desc limit 5;
```

Result Output -

food_id	food_name	Total_Quantity_Sold
9999420	cold drink	19
9999413	chapati	12
9999411	allo paratha	11
9999419	mix paratha	9
9999416	veg biryani	9

7. List the top 3 Zomato employees with the highest average delivery ratings.

```
select employee_id,employee_name,
employee_avg_rating AS Avg_Rating
from zomato_employee
order by Avg_Rating desc limit 3;
```

Result Output -

employee_id	employee_name	Avg_Rating
210123	narayan kumar	4.7
210118	satish p	4.6
210113	jignesh satam	4.5

8. Determine the month with the highest number of total orders placed.

```
select extract(month from order_time) as month_number,
count(order_id) as total_order
from order_detail
group by month_number
order by total_order desc limit 1;
```

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Result Output -

month_number	total_order
10	38

9. Calculate the average order amount for each customer. Order by average order amount in descending order.

```
select customer.customer_id,customer_name ,
avg(order_food.quantity * foods.price_per_unit) as Total_amount
from customer
join order_food on customer.customer_id = order_food.customer_id
join foods  on order_food.food_id = foods.food_id
group by customer.customer_id,customer_name
order by Total_amount desc ;
```

Result Output -

customer_id	customer_name	Total_amount
1009	vighnesh gopinath	320
1013	yogesh jadhav	280
1014	apeksha jadhav	246.666667
1008	punam kale	241
1011	Murtaza khn	197.5
1010	nilesh supekar	181.666667
1007	uday patil	175
1006	sandhya shirale	165
1017	someshwar kale	157.5
1002	pratik kijbil	150
1003	sagar amritkar	130
1001	vaibhav Palkar	127.75
1005	jagruti bhoite	127.5
1004	mitali basargi	112.5
1016	manash konwar	97.5

10. Identify the most frequent customer for each restaurant.

```
with Customer_Frequency as(
select  customer.customer_id,customer_name,restaurant.restaurant_name,
count(order_detail.restaurant_id )  as Most_Vist_Restaurant,
rank() over ( partition by restaurant.restaurant_name
order by count(order_detail.restaurant_id ) desc ) as Restaurant_Rank
from customer
join order_detail on customer.customer_id = order_detail.customer_id
join restaurant  on order_detail.restaurant_id  = restaurant.restaurant_id
group by customer.customer_id,customer_name,restaurant.restaurant_name )
select customer_name, restaurant_name  , Most_Vist_Restaurant from
    Customer_Frequency
     where Restaurant_Rank = 1;
```

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Result Output -

customer_name	restaurant_name	Most_Vist_Restaurant
sandhya shirale	blue water	2
vaibhav Palkar	blue water	2
pratik kijbil	hotel green park	2
punam kale	hydrabadi spice	4
vaibhav Palkar	jalsaa restaurant	3
vaibhav Palkar	laa unico	3
sagar amritkar	saffron	3
nilesh supekar	saffron	3
uday patil	thomson restro	2

11. Calculate the total number of orders placed on weekends.

```
select
dayname(order_time) as Day_name,
count(order_id) as total_order from order_detail
where dayofweek(order_time) in (1,7)
group by Day_Name ;
```

Result Output -

Day_name	total_order
Sunday	6
Saturday	7

12. Calculate the average delivery time (in minutes) for orders placed on weekdays versus weekends.

```
select case when
dayofweek(order_time) in (1,7)then 'weekend'
else 'weekday' end as day_type,
count(order_id) as total_orders,
avg(timestampdiff(minute ,order_time, delivered_time)) as Avg_time
from order_detail
group by day_type ;
```

Result Output -

day_type	total_orders	Avg_time
weekday	30	37.3
weekend	13	37.0769

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13. List the top 5 most expensive food items. Display food_id, food_name, and price_per_unit for all food items with the highest 5 prices.

```
select
   food_id, food_name, price_per_unit as highest_price,  price_rank
from (
   select
      food_id, food_name, price_per_unit,
      dense_rank() over (order by price_per_unit desc) as price_rank
   from foods
) as ranked_foods
where price_rank <= 5
order by highest_price desc;
```

Result Output –

food_id	food_name	highest_price	price_rank
9999416	veg biryani	280	1
9999414	veg thali	220	2
9999418	veg pulav	190	3
9999417	mix veg	180	4
9999419	mix paratha	180	4
9999421	paneer roll	180	4
9999422	veg roll	180	4
9999415	chicken thali	150	5

14. Find the restaurant with the most diverse menu (i.e., the highest number of food items).

```
select r.restaurant_id, r.restaurant_name,
count(distinct (orf.food_id)) as diverse_menu
from restaurant r
join order_food orf on r.restaurant_id = orf.restaurant_id
group by r.restaurant_id, r.restaurant_name
order by  diverse_menu desc limit  1 ;
```

Result Output -

restaurant_id	restaurant_name	diverse_menu
105	laa unico	8

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15. Calculate total payment amount for each payment type.

```
select p.payment_type ,p.payment_status,
count( distinct ofd.order_id) as Total_order,
sum(ofd.quantity * f.price_per_unit) as total_amount
from payment_table p
join order_food ofd on p.order_id = ofd.order_id
join foods f on ofd.food_id = f.food_id
group by p.payment_type ,p.payment_status
order by total_amount desc ;
```

Result Output –

payment_type	payment_status	Total_order	total_amount
internet banking	paid	28	6535
cod	unpaid	15	3370

**********The End **********

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